

Clinical Investigation: Central Nervous System Tumor

Long-term Results of Carbon Ion Radiation Therapy for Locally Advanced or Unfavorably Located Choroidal Melanoma: Usefulness of CT-based 2-Port Orthogonal Therapy for Reducing the Incidence of Neovascular Glaucoma

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Summary

The aim of this study was to determine the long-term results of carbon ion radiation therapy for choroidal melanoma and to assess the usefulness of CT-based, 2-port irradiation in decreasing the risk of neovascular glaucoma. The results were satisfactory in terms of both tumor control and toxicity. It can be concluded that CT-based 2-port irradiation can reduce the high-dose irradiated volume of the iris-

Purpose: To determine the long-term results of carbon ion radiation therapy (C-ion RT) in patients with choroidal melanoma, and to assess the usefulness of CT-based 2-port irradiation in reducing the risk of neovascular glaucoma (NVG).

Methods and Materials: Between January 2001 and February 2012, a total of 116 patients with locally advanced or unfavorably located choroidal melanoma received CT-based C-ion RT. Of these patients, 114 were followed up for more than 6 months and their data analyzed. The numbers of T3 and T2 patients (International Union Against Cancer [UICC], 5th edition) were 106 and 8, respectively. The total dose of C-ion RT varied from 60 to 85 GyE, with each dose given in 5 fractions. Since October 2005, 2-port therapy (51 patients) has been used in an attempt to reduce the risk of NVG. A dose-volume histogram analysis was also performed in 106 patients.

Results: The median follow-up was 4.6 years (range, 0.5-10.6 years). The 5-year overall survival, cause-specific survival, local control, distant metastasis-free survival, and eye retention rates were 80.4% (95% confidence interval 89.0%-71.8%), 82.2% (90.6%-73.8%), 92.8% (98.5%-87.1%), 72.1% (81.9%-62.3%), and 92.8% (98.1%-87.5%), respectively. The overall 5-year NVG incidence rate was 35.9% (25.9%-45.9%) and that of 1-port group and 2-port group were 41.6% (29.3%-54.0%) and 13.9% (3.2%-24.6%) with statistically significant difference ($P < .001$). The dose-volume histogram analysis showed that the average irradiated volume of the iris-ciliary body was significantly lower in the non-NVG group than in the NVG group at

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ciliary body, thereby decreasing the risk of neovascular glaucoma.

all dose levels, and significantly lower in the 2-port group than in the 1-port group at high dose levels.

Conclusions: The long-term results of C-ion RT for choroidal melanoma are satisfactory. CT-based 2-port C-ion RT can be used to reduce the high-dose irradiated volume of the iris-ciliary body and the resulting risk of NVG. © 2013 Elsevier Inc.

Introduction

The outcomes of well-concentrated, high-dose radiation therapy for choroidal melanoma are comparable to those of surgery in terms of tumor control and patient survival (1). In fact, proton therapy has become one of the primary treatments of choice for this type of tumor (2-8), and satisfactory results of stereotactic radiation therapy were also recently reported (9-12). Carbon ion beams can offer even better dose conformity and biological effects than proton beams, although the difference in conformity depends on the depth of the target lesion (13). Therefore, carbon ion radiation therapy (C-ion RT) can be expected to provide outcomes at least comparable to those achieved with proton therapy in the treatment of choroidal melanoma.

At National Institute of Radiological Sciences (NIRS), a phase ½ dose-escalation study of C-ion RT for locally advanced or unfavorably located choroidal melanoma was initiated in 2001, the results of which have already been reported (13). After the phase ½ study, C-ion RT was consecutively performed at the recommended doses: 70 GyE in 5 fractions for T3 tumors and 60 GyE in 5 fractions for T2 or smaller tumors.

Neovascular glaucoma (NVG) is considered to be one of the most serious complications of radiation therapy for choroidal melanoma and thus sometimes leads to enucleation due to severe eye pain. Therefore, it would be useful to develop a method or technique to reduce the risk of severe glaucoma without increasing the risk of recurrence. It has been previously reported by our institute that the irradiated dose and volume of the iris-ciliary body (IC body) and the optic disc are independent risk factors for NVG, based on the results of a dose-volume histogram (DVH) analysis (14). Daftari et al also reported that anterior segment sparing reduces the incidence of complications such as NVG during particle therapy for choroidal melanoma (15). In an attempt to reduce the irradiated volume of the IC body and to decrease the risk of NVG, 2-port orthogonal therapy has been applied to most patients since October 2005.

In this study, the outcomes of C-ion RT were determined using a longer follow-up and more patients than previous reports, and the usefulness of CT-based 2-port orthogonal therapy was assessed with regard to its impact on the risk of NVG.

Methods and Materials

Patients

C-ion RT for choroidal melanoma was initially performed at our institute as part of a phase ½ dose-escalation study for which patients were eligible if they had clinically diagnosed choroidal melanoma with a large tumor (T3 tumor of the International Union Against Cancer [UICC], 5th edition [16]; tumor without extraocular extension, and largest diameter ≥ 15 mm or height ≥ 5 mm) or a tumor close to the optic disk (distance from the optic

disc ≤ 3 mm). After the dose-escalation study was completed, the limitation of tumor size was removed because the carbon ion beams used at the recommended dose were observed to be safe and efficacious. Between January 2001 and February 2012, a total of 116 patients received C-ion RT with CT-based treatment planning. Of these patients, 114 who were followed up for more than 6 months were analyzed for this study.

Patient and tumor characteristics are shown in Table 1. According to the International Union Against Cancer (UICC), 5th edition (16), which was the newest version when the study began, 106 patients (93.0%) had T3 tumors and 8 patients (7.0%) had T2 tumors. In addition, the patients were divided into groups

Table 1 Patient and tumor characteristics of 114 patients with locally advanced or unfavorably located choroidal melanoma received C-ion radiation therapy

Characteristic	n	%
Sex		
Male	59	51.8
Female	55	48.2
Age, y		
Median	56	
Range	22-83	
Tumor size		
Diameter >20 mm and/or height >10 mm	24	21.0
15 mm $<$ diameter ≤ 20 mm and/or 5 mm $<$ height ≤ 10 mm	78	68.4
10 mm $<$ diameter ≤ 15 mm and/or 3 mm $<$ height ≤ 5 mm	12	10.5
T stage (UICC 1997)		
T3	106	93.0
T2	8	7.0
Tumor site (right/left)		
Right	54	47.4
Left	60	52.6
Tumor site (medial/temporal)		
Medial side	49	43.0
Temporal side	64	56.1
Unclassified	1	0.9
Tumor-disc distance		
≤ 3 mm (0)	44 (17)	38.6 (14.9)
>3 mm	70	61.4
Total dose		
60 GyE	16	14.0
70 GyE	78	68.4
77 GyE	13	11.4
85 GyE	7	6.2
No. of ports		
1	63	55.3
2	51	44.7

according to the maximum tumor diameter and height based on Egger's category (2) in which tumor size is categorized as follows: S-size: diameter ≤ 10 mm and height ≤ 3 mm; M-size: $10 \text{ mm} < \text{diameter} \leq 15 \text{ mm}$ and/or $3 \text{ mm} < \text{height} \leq 5 \text{ mm}$; L-size: $15 \text{ mm} < \text{diameter} \leq 20 \text{ mm}$ and/or $5 \text{ mm} < \text{height} \leq 10 \text{ mm}$; and XL-size: diameter $> 20 \text{ mm}$ and/or height $> 10 \text{ mm}$. The numbers of patients with S-size, M-size, L-size, and XL-size tumors were 0, 12 (10.5%), 78 (68.4%), and 24 (21.0%), respectively. Regarding the tumor site in the globe, the distance from the tumor to the optic disk (tumor-disc distance) was $\leq 3 \text{ mm}$ in 44 patients (38.6%). Among these cases, 17 tumors (14.9%) were attached to the optic disk.

This study was approved by the Institutional Review Board of NIRS. Written informed consent was obtained from each patient.

Treatment planning and irradiation techniques

The details of the treatment procedure have been previously reported (13, 14). CT-based treatment planning was performed in each case, initially using our customized planning system, EYEPLAN (NIRS, Japan). After April 2005, the planning system was changed to a commercial radiation therapy treatment planning system, XiO-carbon combined with FocalPro (CMS Inc, St. Louis, MO).

The total prescribed dose in the phase $\frac{1}{2}$ study was determined referring to previously reported studies of proton therapy (2-8, 17), as the relative biological effectiveness of carbon ion beams in choroidal melanoma was unknown, a dose-escalation study was performed. The actual irradiated dose varied from 60 GyE to 85 GyE: 16 patients (14.0%) received 60 GyE, 78 patients (68.4%) received 70 GyE, 13 patients (11.4%) received 77 GyE and 7 patients (6.2%) received 85 GyE. All patients were treated daily in 5 fractions. The doses used in the prior 42 patients (36.8%) who received C-ion RT before February 2004 were selected according to the dose-escalation protocol, and the doses used in the subsequent patients were selected according to the recommended doses obtained in the study: 70 GyE in 5 fractions for T3 tumors and 60 GyE in 5 fractions for tumors T2 or smaller.

Until September 2005, C-ion RT was performed using a single anterior port only. The use of 2-port orthogonal therapy was initiated in October 2005 and has been applied in 51 patients (44.7%) thus far. The typical dose distributions of both treatment techniques are shown in Figure 1a and b.

After treatment, the patients were followed by both ophthalmologists and radiation oncologists. NVG was defined as an intraocular pressure greater than 21 mm Hg in the presence of iris neovascularization.

DVH analysis

A DVH analysis was performed in 106 patients to assess the relationship between the DVH of the IC body and the occurrence of NVG. In the remaining 8 patients treated in the 2-port group, the DVH of the IC body could not be accurately calculated because 2 different series of CT images with different gazing point directions were used for treatment planning.

The IC body was defined as the expanding volume of the lens measured 3 mm laterally, adding the volume of the anterior chamber and subtracting the lens volume. An example is shown in Figure 1c. This volume was not treated as an organ at risk or dose constraint at the time of planning, but rather was taken into account in determining the number of ports.

The average DVHs of the patients with and those without NVG were compared, and the differences were assessed using the Student *t* test. The threshold of the V_{50} (volume irradiated ≥ 50 GyE) of the IC body was selected to be 0.1 mL, according to our previous report, in which a value of 0.127 mL was shown to be optimal (14).

Statistical analysis

Estimates of the overall survival (OS), cause-specific survival (CSS), local control rate (LCR), distant metastasis-free survival rate (DMFSR), eye retention rate (ERR), and NVG incidence were calculated according to the Kaplan-Meier method. The outcomes were assessed from the first day of C-ion RT to the date of an event or the last follow-up. The log-rank test was used for the univariate analysis, and a Cox proportional hazards model was used for the multivariate analysis to identify any predictors.

The threshold of the tumor-disc distance was selected to be 3 mm when determining the incidence of NVG because the planning target volume was defined as at least a 2.5-mm margin added to the gross tumor volume, making it difficult to avoid the optic disc if the tumor-disc distance was less than 3 mm.

The results were considered to be significant at a *P* value of $< .05$. All statistical analyses were performed using the SPSS software program (SPSS Inc, Chicago, IL).

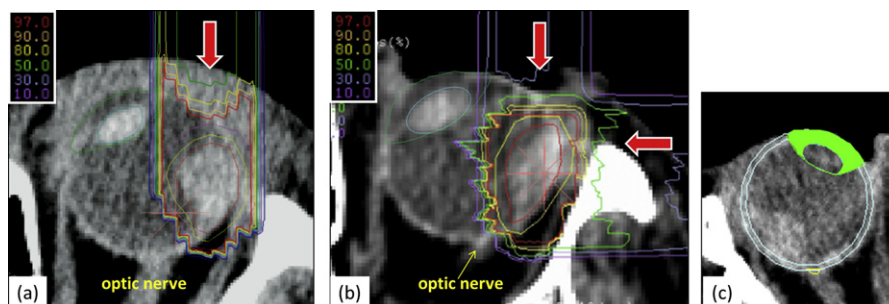


Fig. 1. (a and b) Dose distributions of the 1-port and 2-port therapies. Inside red line indicates the gross tumor volume and yellow line indicates the clinical target volume. Left upper box shows the percentage of the relative dose. (c) Green-filled area indicates the iris-ciliary body.

Results

Summarized results are shown in Tables 2 and 3. The median duration of follow-up in the 114 analyzed patients was 4.6 years (range, 0.5-10.6 years). Thus far, 21 patients have died, 19 of whom died of choroidal melanoma. Local recurrence developed in 6 patients. The areas of recurrence were within the irradiated volume in 2 patients (the dose used in 1 patient was 70 GyE, and that used in the other was 77 GyE) and in another part of the treated eye in the other 4 patients. Five of the patients with recurrence underwent secondary enucleation and 1 patient underwent reirradiation with C-ion RT. Distant metastasis occurred in 28 patients, with liver, lung, bone, and meningeal dissemination observed in 19, 6, 2 and 1 patients, respectively. The 5-year OS, CSS, LCR, and DMFSR were 80.4% (95% confidence interval 89.0-71.8%), 82.2% (90.6-73.8%), 92.8% (98.5-87.1%), and 72.1% (81.9-62.3%), respectively. In the univariate and multivariate analyses, tumor size was not found to be significant with respect to oncological outcome.

NVG developed in 36 patients, among 30 and 6 patients in the 1-port and 2-port groups, respectively. The 3-year and 5-year NVG incidence rates were 29.7% (20.9-38.5%) and 35.9% (25.9-45.9%), respectively. The tumor size, tumor-disc distance (0-3 mm vs others), and number of ports were found to be statistically significant predictors of the incidence of NVG in the multivariate analysis, with *P* values of <.001, .031, and .005, respectively. Concerning the number of ports, the 3-year NVG incidence rates were 41.6% (29.3-54.0%) and 13.9% (3.2-24.6%) for the 1-port and 2-port group, respectively (Fig. 2a). There was a statistically significant difference between the 2 groups (*P*<.001).

The results of the DVH analysis are shown in Figure 3. The average irradiated volume of the IC body was significantly lower in the non-NVG group than in the NVG group at all dose levels, and significantly lower in the 2-port group than in the 1-port group at high dose levels. Concerning the V_{50} of the IC body, the 3-year incidences of NVG were 60.9% (45.1-76.7%) and 11.4% (3.3-19.5%) in the $V_{50} > 0.1$ mL and $V_{50} \leq 0.1$ mL groups, respectively (Fig. 2b). There was a statistically significant difference between the 2 groups (*P* value <.001).

Enucleation was required in 8 patients because of local recurrence (5 patients) and severe glaucoma (3 patients in the 1-port group; the dose used in 1 patient was 77 GyE and in the other 2 patients was 60 GyE). No patients in the 2-port group underwent enucleation because of severe glaucoma. The 5-year ERR was 92.8% (98.1-87.5%). With respect to the number of ports, the 3-year ERR was 90.1% (97.6-82.6) and 96.1% (100-89.9%) for the 1-port and 2-port groups, respectively. No statistically significant differences were observed between the 2 groups. No dose dependency was observed.

A visual acuity $\geq 20/200$ was preserved in 43 of 78 patients (55.1%) with an initial visual acuity $\geq 20/200$. No dose dependency was found.

Discussion

In this study, the OS, LCR, DMFSR, and ERR at 5 years were 80.4%, 92.8%, 72.1%, and 92.8%, respectively. The rates observed in the patients with large tumors corresponded to previously published rates, with OS, LCR, DMFSR, and ERR at 5

Table 2 Summary of the long term results of C-ion radiation therapy for choroidal melanoma

	3 y (%)	5 y (%)	<i>P</i> value
Overall survival (all patients)	88.4	80.4	
Tumor-disc distance			
0	74.0	53.8	.004
Others	91.3	86.2	
Cause-specific survival (all patients)	90.5	82.2	
Tumor-disc distance			
0	74.0	58.3	.009
Others	91.3	87.3	
Local control rate (all patients)	95.7	92.8	
Distant metastasis-free survival rate (all patients)	84.6	72.1	
Tumor-disc distance			
0	60.3	37.7	.002
Others	89.1	78.8	
Eye retention rate (all patients)	94.1	92.8	
NVG incidence (all patients)	29.7	35.9	
Tumor size			
Diameter >20 mm and/or height >10 mm	64.4	64.4	<.001
Others	20.9	20.9	
Tumor-disc distance			
≤ 3 mm	55.1	51.4	.004
>3 mm	80.1	72.2	
No. of ports			
1	41.6	49.0	<.001
2	13.9	13.9	
DVH parameter			
$V_{50} > 0.1$ mL	60.9	65.4	<.001
$V_{50} \leq 0.1$ mL	11.4	18.7	

Abbreviations: NVG = neovascular glaucoma; DVH = dose-volume histogram; V_{50} = volume irradiated ≥ 50 GyE.

years ranging from 70% to 85%, 84% to 96%, 76% to 88%, and 78% to 93%, respectively (2-9, 11).

Tumor size was not a significant factor predicting the OS, CSS, LCR, or DMFSR in this study, although both tumor size and T stage have been found to be significant prognostic factors in other reports (2, 4-8). The reason for this discrepancy may be that the majority of the tumors observed in our study were large, such as T3 (93%) and L/XL-size (79%) tumors.

Our previous report using a DVH analysis with CT-based planning indicated that the V_{50} of the IC body and optic disc irradiation were independent risk factors for NVG, and new techniques to reduce the dose to the anterior segment and optic disc were expected to decrease the incidence of NVG (14). Based on that report, we began to use 2-port therapy in an attempt to reduce the risk of NVG, and 51 patients have undergone C-ion RT with 2 ports since October 2005.

In the present study, the 3-year incidence rate of NVG in the 2-port group was 13.9%, which was decreased compared to that observed in the 1-port group (41.6%) and was much lower than that reported in other studies (24-38%) (4, 8, 10, 11). Moreover, no patient in the 2-port group underwent enucleation because of severe glaucoma, whereas 3 of the 63 patients in the 1-port group underwent enucleation because of severe glaucoma.

Table 3 Statistical analysis of study results

	Overall survival	Cause-specific survival	Local control rate	Distant metastasis-free survival rate	Eye retention rate	Neovascular glaucoma
Univariate analysis (log-rank test)						
Sex						
Male/female	<u>.048</u>	<u>.013</u>	.52	<u>.014</u>	.85	.49
Age, y						
<50/≥50	.67	.93	.52	.31	.96	.37
Tumor size						
XL size/others	.28	.43	.85	.54	.16	<u><.001</u>
Tumor site						
Medial side/temporal side	<u>.028</u>	<u>.024</u>	.677	<u>.042</u>	.27	.74
Tumor-disc distance						
0/others	<u>.004</u>	<u>.009</u>	.31	<u>.002</u>	.91	<u>.004*</u>
No. of ports						
1 port/2 ports	.090	.15	.41	.10	.18	<u><.001</u>
Multivariate analysis (Cox proportional hazards model)						
Sex						
Male/female	.11	.072	.54	.081	.82	.84
Age, y						
<50/≥50	.46	.72	.53	.18	.94	.20
Tumor size						
XL-size/others	.32	.47	.86	.56	.14	<u><.001</u>
Tumor site						
Medial side/temporal side	.055	<u>.047</u>	.68	.078	.27	.58
Tumor-disc distance						
0/others	<u>.013</u>	<u>.024</u>	.30	<u>.003</u>	.90	<u>.031*</u>
No. of ports						
1port/2 ports	.42	.59	.43	.33	.19	<u>.005</u>

Abbreviations: XL-size: diameter >20 mm and/or height >10 mm.

Underlined data signify *P* value <.05.

* ≤3 mm/others.

A multivariate analysis identified tumor size, tumor-disc distance, and use of 1-port therapy to be significant risk factors for NVG. The DVH analysis showed that the average irradiated volume of the IC body was significantly lower in the non-NVG and 2-port groups than in the NVG and 1-port groups, whereas the incidence of NVG was significantly lower in the lower V_{50} group.

It is thought that carbon ion beams exhibit superior dose conformity in terms of their sharp lateral fall off compared with proton and photon beams, so that there is less of an effect on the

surrounding normal tissue (13). Nevertheless, if the tumor-disc distance is less than 3 mm, it is difficult to avoid the optic disc, and therefore the risk of NVG is considered to increase because of damage to the central retinal vessels.

With respect to the mechanisms underlying the development of NVG associated with choroidal melanoma, Lee et al suggested that possible causes of anterior ocular segment and retinal neovascularization include the following: release of angiogenic factors; serous retinal detachment with secondary retinal

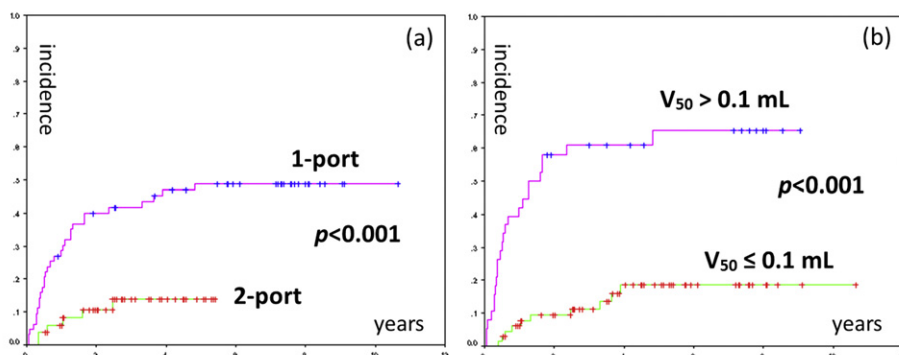


Fig. 2. Incidences of neovascular glaucoma (NVG) in the 2-port (a) and lower V_{50} (b) groups were significantly lower than those of the other groups in the univariate analysis.

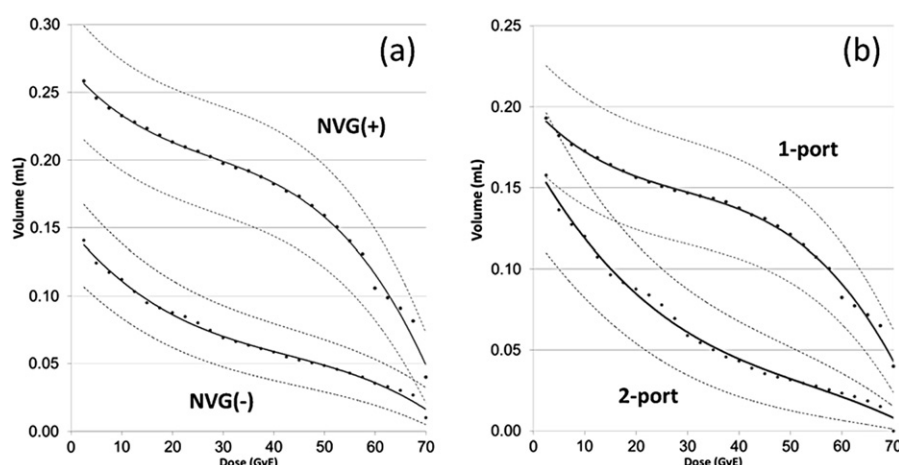


Fig. 3. Dose-volume histogram (DVH) analysis was performed in 106 of the 114 patients. Differences between the 2 groups were assessed using Student *t* test at each dose level. Solid lines indicate approximated curve of the average volume of each dose level, and dotted lines indicate that of the 95% confidence interval. (a) Average irradiated volume of the iris-ciliary body was significantly lower in the non-neovascular glaucoma (NVG) group than in the NVG group at all dose levels. (b) Average irradiated volume of the iris-ciliary body was significantly lower in the 2-port group than in the 1-port group at high dose levels.

ischemia; retinal vascular occlusion secondary to retinal invasion by the tumor; inflammation; and retinopathy after radiation therapy (18). Regarding the impact of the V_{50} of the IC body on the occurrence of NVG, a large irradiated volume in the anterior segment of the eye may lead to local ischemic changes or severe inflammation, possibly inducing neovascularization. On the other hand, the tumor-disc distance was also found to be a significant predictive factor for the development of NVG after C-ion RT in this analysis. Fernandes et al reported that the risk of NVG for juxtapapillary tumors results from radiation damage to the posterior chamber rather than damage to the anterior chamber based on the correlation of dosimetric data with the pathological findings of 10 eyes enucleated after stereotactic radiation therapy (12). A likely mechanism is that disturbance of the circulatory system by high-dose irradiation of the central retinal vessels induces neovascularization after radiation therapy. Recently, the intravitreal injection of bevacizumab, an antivascular endothelial growth factor humanized monoclonal antibody, was suggested to be useful for the treatment of NVG because neovascularization may play an important role in the pathogenesis of NVG. In addition, the antibody may be useful for preventing NVG in high-risk patients who have received C-ion RT (19).

This study confirmed the results of our previous report regarding the effectiveness of CT planning and a DVH analysis of the IC body (14). To the best of our knowledge, CT-based planning of particle radiation therapy for this type of tumor has never been reported from other institutes, and Jeganathan et al reported that there is little useful fractionation in the literature regarding the relationship between fractionation and volume effects on the development of NVG (20).

There was no significant differences in the incidence of NVG between the 50 Gy and 70 Gy groups in the randomized controlled trial of Gragoudas et al (17). Therefore, it is difficult to explain the relationship between the risk of NVG and the radiation dose. However, our DVH analysis clarified the volume effects of the IC body on the incidence of NVG.

This study has several limitations. First, the number of patients included in this analysis was relatively small. However, choroidal melanoma is a rare disease in our country, with only about 20 to

30 cases per year among the total national population (13). Second, the analysis performed in this study was retrospective and lacked randomization, although the original phase Ⅱ study was prospective. It may be necessary to conduct a prospective study to better clarify the efficacy of 2-port orthogonal therapy. Conducting a large prospective trial using a quantitative analysis method, such as a DVH analysis, is recommended to evaluate the efficacy and safety of multi-port proton therapy and to provide radiation oncologists more information regarding dose constraints in organs at risk, even with proton or photon radiation therapy.

Conclusion

In conclusion, CT-based 2-port C-ion RT is useful for reducing the high-dose irradiated volume of the IC-body and the risk of NVG. This study also confirms the efficacy of CT planning and DVH analyses of the IC body. Our long-term results regarding the use of C-ion RT to treat choroidal melanoma are satisfactory, even in patients with large tumors or tumors in unfavorable sites. This study suggests that the tumor size, the tumor-disc distance and the number of ports are all related to the occurrence of NVG.

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